

Matthew I. Swindall, Ph.D.

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Professional Profile

Computational scientist with expertise in artificial intelligence, machine learning, and scientific software development across interdisciplinary research domains. Current work focuses on AR/VR/XR development, cybersecurity, nuclear forensics and safeguards, radiochemistry, laboratory automation, and multimodal knowledge base design for GraphRAG applications. Previous research includes applied AI methodologies in computer vision, handwritten text recognition (HTR/OCR), natural language processing, and statistical analysis. Creator of AL-PUB, a large-scale crowdsourced image dataset of ancient Greek papyrus characters for computer vision research.

Research Areas

Applied AI/ML, computer vision, handwritten text recognition, NLP, multimodal knowledge systems, GraphRAG, AR/VR/XR, cybersecurity, nuclear forensics and safeguards, radiochemistry, laboratory automation, scientific software development, robotics, blockchain, and scientific computing.

Education

Ph.D., Computational Science — Middle Tennessee State University	2024
M.S., Computer Science — Middle Tennessee State University	2024
B.S., Physics — Middle Tennessee State University	2016
A.S., Physics — Columbia State Community College	2016

Technical Skills

AI/ML Concepts: Deep learning, neural networks, convolutional neural networks, recurrent neural networks, generative adversarial networks, reinforcement learning, transfer learning, autoencoders, transformers, LLMs, RAG, GraphRAG

AI/ML Frameworks and Libraries: TensorFlow, PyTorch, Keras, scikit-learn, spaCy, BERTopic, NLTK, OpenCV, Ollama

Programming Languages: Python, C/C++, R, JavaScript, HTML, CSS, SQL, Solidity, Cypher, Swift, SwiftUI, C#, ColdFusion

Python Libraries: NumPy, Pandas, SciPy, Matplotlib, OpenCV, scikit-learn, Tkinter, Flask

Computing and Software: Unix/Linux, Windows, iOS, parallel processing, cluster computing (SLURM), Neo4j, Unity, Blender, Reality Composer Pro, Igor Pro, FreeCAD, DaVinci Resolve

Technical Domains: AR/VR/XR, AI/ML, graph databases, blockchain and smart contracts, laboratory automation, robotics, IoT, Raspberry Pi, Arduino, 3D printing, CAD, CNC, visionOS

Scientific Background: Physics, astronomy, chemistry, engineering, electronics, linear algebra, calculus, differential equations, probability, statistics

Professional Experience

Postdoctoral Appointee

Jan 2025 – Present

Argonne National Laboratory

- Contribute to multiple nuclear forensics and safeguards/nonproliferation projects within the Strategic Security Sciences Division.
- Develop spatial computing applications (AR/VR/XR) for national security-aligned research and operational use cases.
- Build software tools related to national security, cybersecurity defense, and microfluidics hardware operation.
- Design, prototype, and test 3D components for radiological and X-ray imaging experiments, including chemical flow cells and containment hardware.
- Conduct interdisciplinary AI/ML research and contribute to technical reports, proposals, and peer-reviewed publications.

Graduate Assistant

Sep 2019 – May 2024

Middle Tennessee State University

- Led and supported computational projects involving optical character recognition, object detection and recognition, semantic segmentation, microcontrollers, microcomputers, electrical circuits, robotics, 3D printing, laser etching/cutting, and augmented/virtual reality.
- Designed and prototyped STEM demonstrations and institutional hardware/software tools.
- Provided equipment training and technology coaching.
- Designed and curated STEM exhibits, including a Lego block sorting robot, a Bluetooth-enabled LED text sign with Android app integration, and a real-time object detection and segmentation exhibit.

Physics and Astronomy Tutor

Jan 2015 – Dec 2016

Middle Tennessee State University

- Tutored undergraduate students in calculus-based physics, electricity and magnetism, classical mechanics, introductory astronomy, solar system astronomy, stars/galaxies/cosmology, first-year chemistry, and calculus.

Selected Projects and Research Contributions

Microfluidics and Laboratory Automation

Led software and hardware development and laboratory automation workflows supporting microfluidics operation for small-angle X-ray scattering studies of lanthanide and actinide precipitation at the Advanced Photon Source.

AR/VR/XR for National Security Applications

Developed spatial computing software aligned with interdisciplinary research and national security use cases including nuclear forensics and nonproliferation.

Blockchain for Digital Edition Management

Developed a blockchain-based framework for smart digital edition management in papyrology.

AI-Assisted Papyrology

Contributed to research and platform development integrating deep learning into papyrological work-

flows, including OCR, dating, and digital scholarly tools.

AL-PUB Dataset

Created a large-scale crowdsourced image dataset of ancient Greek papyrus characters for computer vision tasks.

Grants and Funding

Department of Energy LDRD Innovate SEED — \$25,000

2026

Publications

Conference Proceedings and Workshop Papers

- Blakely, B., Ramchandra, A. R., **Swindall, M.**, Fenu, S., Balaji, S. R. A., Jhala, K., Plathottam, S., & Ranganathan, P. (2026). A pipeline-ordered diagnostic protocol for online reinforcement learning with game-theoretic defense and delayed environmental reward in cyber-physical systems. In *IEEE Cyber Awareness & Research Symposium (CARS)*.
- West, G., **Swindall, M. I.**, Brusuelas, J. H., Maltomini, F., Gerhardt, M., D’Angelo, M., & Wallin, J. F. (2024). A deep learning pipeline for the palaeographical dating of ancient Greek papyrus fragments. In *Machine Learning for Ancient Languages (ML4AL) Workshop at the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*.
- Swindall, M. I.**, Upadhyay, K., Brusuelas, J. H., West, G., & Wallin, J. F. (2024). Smart digital edition management: A blockchain framework for papyrology. In *Proceedings of the 2024 Computers and People Research Conference*. doi:10.1145/3632634.3655860
- Swindall, M. I.**, West, G., Brusuelas, J. H., Williams, A. C., & Wallin, J. F. (2024). Towards a platform for AI-assisted papyrology. In *Past Meets Future Workshop at the 2024 International Conference on Intelligent User Interfaces (IUI 2024)*.
- Swindall, M. I.**, West, G., Brusuelas, J. H., & Wallin, J. F. (2022). Crowdsourcing image datasets: An examination of ground-truth in labeling, text segmentation, and sampling bias. In *The Tenth AAAI Conference on Human Computation and Crowdsourcing Doctoral Consortium*.
- Swindall, M.**, Player, T., Keener, B., Williams, A., Brusuelas, J., Nicolardi, F., D’Angelo, M., Vergara, C., McOske, M., & Wallin, J. (2022). Dataset augmentation in papyrology with generative models: A study of synthetic ancient Greek character images. In *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence (Vol. 2022, pp. 4973-4979)*. IJCAI
- Swindall, M. I.**, Croisdale, G., Hunter, C. C., Keener, B., Williams, A. C., Brusuelas, J. H., Krevans, N., Sellew, M., Fortson, L., & Wallin, J. F. (2021). Exploring learning approaches for ancient Greek character recognition with citizen science data. In *2021 17th International Conference on eScience*, 128–137.

Journal Articles

- West, G., **Swindall, M. I.**, Keener, B., Player, T., Williams, A. C., Brusuelas, J. H., & Wallin, J. F. (2024). Incorporating crowdsourced annotator distributions into ensemble modeling to improve classification trustworthiness for ancient Greek papyri. *Journal of Data Mining & Digital Humanities, Documents historiques et reconnaissance automatique de texte*. jdmdh.episciences.org/12958

Technical Reports

- Blakely, B., Ramchandra, A. R., **Swindall, M.**, Fenu, S., & Balaji, S. R. A. (2026). Adaptive Cybersecurity for Distributed Energy Resources (AdCyDER): Online reinforcement learning with Stackelberg-optimized defenses. doi:10.2172/3377852

Ph.D. Dissertation

Advancing Digital Papyrology: Machine Learning and Blockchain Tools for Modernizing the Study of Ancient Greek Manuscripts. Matthew I. Swindall. Middle Tennessee State University, 2024.

Dissertation Committee: Dr. John F. Wallin (Advisor), Dr. James H. Brusuelas, Dr. Joshua L. Phillips, Dr. Qiang Wu

Invited Talks

Understanding Ancient Manuscripts Using Crowd-sourcing and Data Science. Matthew I. Swindall, Graham West, James H. Brusuelas, and John F. Wallin. AWS Human-in-the-Loop Science Summer Seminar Series. Virtual. Aug. 17, 2023.

Generative AI and Higher Education. John F. Wallin, Matthew I. Swindall, and Isaac Shirk. Middle Tennessee State University AI Initiative Research Conference. Murfreesboro, Tennessee. Nov. 10, 2023.

A.I. Assisted Papyrology: Integrating Deep Learning into the Scholarly Workflow. Matthew I. Swindall, Graham West, James H. Brusuelas, and John F. Wallin. Alpha, Aleph, and AI: Languages of the Ancient Mediterranean and Near East Conference. Bristol, United Kingdom. June 14, 2023.

Dataset Augmentation in Papyrology with Generative Models: A Study of Synthetic Ancient Greek Character Images. Matthew I. Swindall. Middle Tennessee State University Scholars Day. Murfreesboro, Tennessee. Mar. 22, 2022.

Professional Service and Outreach

- Instructor, Coding for Science Camp, Argonne National Laboratory, 2025 and 2026
- Engagement Officer, Postdoctoral Society of Argonne, 2026
- Member-at-Large (SSS), Postdoc Symposium Committee, and Postdoc Awareness Week Committee, 2025
- Program Committee, Machine Learning for Ancient Languages (ML4AL) Workshop at ACL 2024
- Reviewer, ACM SIGMIS Computers and People Research (CPR), 2024

Internships and Honors

La Serena School for Data Science	2021
NSF and CMM/UCHile-funded Data Science for Astronomy Program in partnership with NOIRLab/AURA	
MTeach Summer Internship	2016
STEM teaching internship in partnership with the Discovery Center at Murfree Spring	

Certifications

Neo4j Graph Database Certified Professional